Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title MK-5684-01A Substudy: A Phase 1/2 Umbrella Substudy of MK-5684-U01 Master Protocol to Evaluate the Safety and Efficacy of MK-5684-based Treatment Combinations or MK-5684 Alone in Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC) **Short Title/Acronym** Substudy 01A: Safety and Efficacy of Opevesostat (MK-5684)-Based Treatment Combinations or Opevesostat Alone **Protocol Number** MK-5684-01A (5684-01A) **NCT Number** NCT06353386 **Sponsor Name** Merck **Trial Status** Recruiting **Investigational Product** Opevesostat (MK-5684) alone or in combination (olaparib, docetaxel, cabazitaxel) **Phase of Development** Phase 1 / Phase 2 **Medical Monitor** Clinical Operations Lead, Merck

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: *Safety Lead-in Phase:* To evaluate safety and tolerability, and to establish a recommended Phase 2 dose (RP2D) for the opevesostat-based treatment combinations. There will be no hypothesis testing in this study. *Efficacy Phase:* To evaluate safety and efficacy of opevesostat-based combinations or as a single agent in participants with mCRPC. **Secondary Objectives:** To evaluate objective response rate (ORR), radiographic progression-free survival (rPFS), and overall survival (OS).

2.2 Background & Rationale Current treatments for mCRPC are associated with hormone dependence and the development of resistance. Therapeutic agents with novel mechanisms of action are needed for this patient population. Opevesostat (MK-5684; ODM-208) is an oral, nonsteroidal inhibitor of cytochrome P450 11A1 (CYP11A1), a catalyst of the first and rate-limiting step of steroid biosynthesis. In previous phase 1/2 studies, opevesostat demonstrated significant antitumor activity in patients with heavily pretreated mCRPC.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Substudy) **Control Type:** Active Comparator (Opevesostat alone vs. combination arms) **Blinding:** Open-label **Randomization:** 1:1:1:1 (Efficacy Phase) stratified by AR-LBD mutation status

3.2 Planned Enrollment & Duration Target N: 220 participants **Number of Sites:** Multicenter (Global, including sites in the US such as UC San Diego and UCLA) **Estimated Study Start:** 20-May-2024 **Estimated Primary Completion:** 31-Mar-2028

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of adenocarcinoma of the prostate without small cell histology. 2. Prostate cancer progression and received androgen deprivation therapy (ADT) or post bilateral orchiectomy within 6 months before screening. 3. Evidence of disease progression from either >4 weeks from last flutamide treatment, or >6 weeks from last bicalutamide or nilutamide treatment (if receiving first generation anti-androgen therapy as last treatment). 4. Current evidence of metastatic disease. 5. Prior treatment with 1 to 2 novel hormonal agent(s) (NHA) for non-metastatic, or metastatic, hormone-sensitive prostate cancer or castration-resistant prostate cancer and have disease progression during or after treatment. 6. Treatment with bone resorptive therapy (including bisphosphonate or denosumab) must have been on stable doses for >4 weeks before randomization. 7. Participants who experienced adverse events (AEs) due to previous anticancer therapies must have recovered to <Grade 1 or baseline. 8. HIV-infected participants must have well-controlled HIV on antiretroviral therapy. HBsAg positive participants are eligible if receiving HBV antiviral therapy for ≥ 4 weeks with undetectable viral load. Participants with a history of HCV are eligible if HCV viral load is undetectable.

4.2 Exclusion Criteria 1. History of pituitary dysfunction or poorly controlled diabetes mellitus. 2. Active or unstable cardio/cerebro-vascular disease, including thromboembolic events, history of stroke/TIA within 6 months, MI within 6 months, NYHA Class III or IV, symptomatic coronary heart disease, or unstable angina. 3. History or family history of long corrected QT interval (QTc) syndrome. 4. Myelodysplastic syndrome (MDS) / acute myeloid leukemia (AML) or features suggestive of MDS/AML. 5. History or current condition of adrenal insufficiency. 6. History of (noninfectious) pneumonitis requiring steroids, or current pneumonitis. 7. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 8. Undergone major surgery within 28 days before randomization without recovery from toxicities/complications. 9. On an unstable dose of thyroid hormone therapy within 6 months prior to the first dose. 10. Received a whole blood transfusion in the last 120 days before randomization. 11. Received prior systemic anticancer therapy within 4 weeks before randomization, or prior radiotherapy within 2 weeks of starting study intervention. 12. Received a live or live-attenuated vaccine within 30 days before the first dose (killed vaccines allowed). 13. Diagnosis of immunodeficiency, or receiving chronic systemic steroid/ immunosuppressive therapy within 7 days prior to the first dose. 14. Known additional malignancy progressing or requiring active treatment within the past 3 years. 15. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 16. Active autoimmune disease requiring systemic treatment in the past 2 years. 17. Active infection requiring systemic therapy. 18. Concurrent active HBV or HCV infections.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Opevesostat 5 mg PO BID alone, or Opevesostat 5 mg PO BID plus olaparib (RP2D), Opevesostat 5 mg PO BID plus docetaxel (RP2D), or Opevesostat 5 mg PO BID plus cabazitaxel (RP2D). **Route of Administration:** Oral (Opevesostat, Olaparib) and Intravenous (Docetaxel, Cabazitaxel). **Duration of Treatment:** Continuous until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Bone resorptive therapy (e.g., bisphosphonate or denosumab) if on stable doses for >4 weeks. Administration of killed vaccines is allowed. **Prohibited:** Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Active autoimmune disease requiring systemic treatment in the past 2 years.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Imaging, PSA/Labs | | **Baseline** | Day 0 | Randomization, First Dose, Safety Assessment | | **Interim** | Every 3-4 Weeks | Safety Labs, Efficacy Assessment (PSA levels), Tumor Imaging | | **End of Study** | Variable | Final Vital Signs, Exit Interview, IP Accountability, rPFS confirmation |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Number of participants who experience one or more dose-limiting toxicities (DLTs):** Includes Grade 4 nonhematologic toxicity, Grade 4 hematologic toxicity lasting >7 days, any nonhematologic AE >Grade 3, Grade 3/4 febrile neutropenia, prolonged delay (>2 weeks) in initiating treatment, missing >25% of doses due to drug-related AEs, or Grade 5 toxicity. * **Number of participants who experience one or more adverse events (AEs).** * **Number of participants who discontinue study intervention due to an AE.**

7.2 Secondary Endpoints * Objective response rate (ORR). * Radiographic progression-free survival (rPFS). * Overall survival (OS).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard oncology Phase 1/2 AE collection via patient interviews, physical exams, and lab assessments prior to each cycle. **Serious Adverse Events (SAEs):** Required reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal and external safety review committees to evaluate dose-limiting toxicities (DLTs) during the safety lead-in phase.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints. Safety Set for all participants who received at least one dose of study treatment. **Sample Size Justification:** 220 subjects. Sample size determined to adequately evaluate safety and establish RP2D and provide sufficient power to detect a clinically meaningful response in the efficacy phase. **Hypothesis Testing:** There will be no hypothesis testing in this study. **Handling of Missing Data:** Standard oncology guidelines for RECIST v1.1 and PCWG3 criteria (e.g., censoring at last evaluable disease assessment).

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits based on the risk-based monitoring plan. **Audit Trail:** Complete electronic Case Report Form (eCRF) audit trails, imaging central reads by Blinded Independent Central Review (BICR), and data clarification mechanisms.

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11. Study Identifiers & Governance

Primary ID: {{MK-5684-01A}} **Registry Number:** {{NCT06353386}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{Substudy 01A: Safety and Efficacy of Opevesostat (MK-5684)-Based Treatment Combinations or Opevesostat Alone}} **Official Regulatory Title:** {{MK-5684-01A Substudy: A Phase 1/2 Umbrella Substudy of MK-5684-U01 Master Protocol to Evaluate the Safety and Efficacy of MK-5684-based Treatment Combinations or MK-5684 Alone in Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC)}}

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12. Clinical Framework (mCRPC Specific)

Primary Condition: {{Prostatic Neoplasms, Castration-Resistant}} **Diagnosis Threshold:** {{Progression during ADT and on/after 1-2 NHAs}} **Medical Methodology:** {{Opevesostat alone or in combination with olaparib/taxanes}} **Optimization Target:** {{Evaluating safety, tolerability, establishing RP2D, and efficacy outcomes}}

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13. Study Procedures & Methodology

Management Pathway: {{Oncology standard of care for mCRPC}} **Education Protocol (HCP):** {{Protocol training and RP2D administration guidelines}} **Education Protocol (Patient):** {{Informed consent, dosing diaries, AE reporting}} **Quality Audit Mechanism:** {{Central imaging review (BICR) and safety monitoring}}

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14. Observation & Follow-Up Schedule

Total Duration: {{Estimated ~4 years total study length}} **Onsite Visit Cadence:** {{Every 3-4 weeks per treatment cycle}} **Remote Monitoring Frequency:** {{Continuous AE monitoring}} **Checklist Review Interval:** {{Per cycle assessment}}

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]